Nanocloud01

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Executive Summary

Project Team

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Optimization Results

Post-main epoch (PPL 21.61), careful low-LR fine-tuning and checkpoint selection achieved superior results.

Best Validation Perplexity

18.7681

Architecture & Strategy

GPT-style <100M Params

- Modern Components: RMSNorm, RoPE, SwiGLU, QK-Norm
- Optimizer: Muon/AdamW Split
- Mixed-domain Training Data

Key Learnings

- Data mixture and late-stage LR scheduling are more impactful than complex architecture changes or distillation.
- Intermediate checkpoint evaluation is critical; the final checkpoint is not always the highest performing.
- Deeper over Wider for limited Parameters

Data Pipeline & Model Architecture

Data Pipeline & Strategy

Tokenizer: GPT-2 BPE, uint16 binary shards for high-speed streaming.

Corpus: Shifted to curated multi-domain: FineWeb-Edu, Wikipedia, academic texts, and literary works.

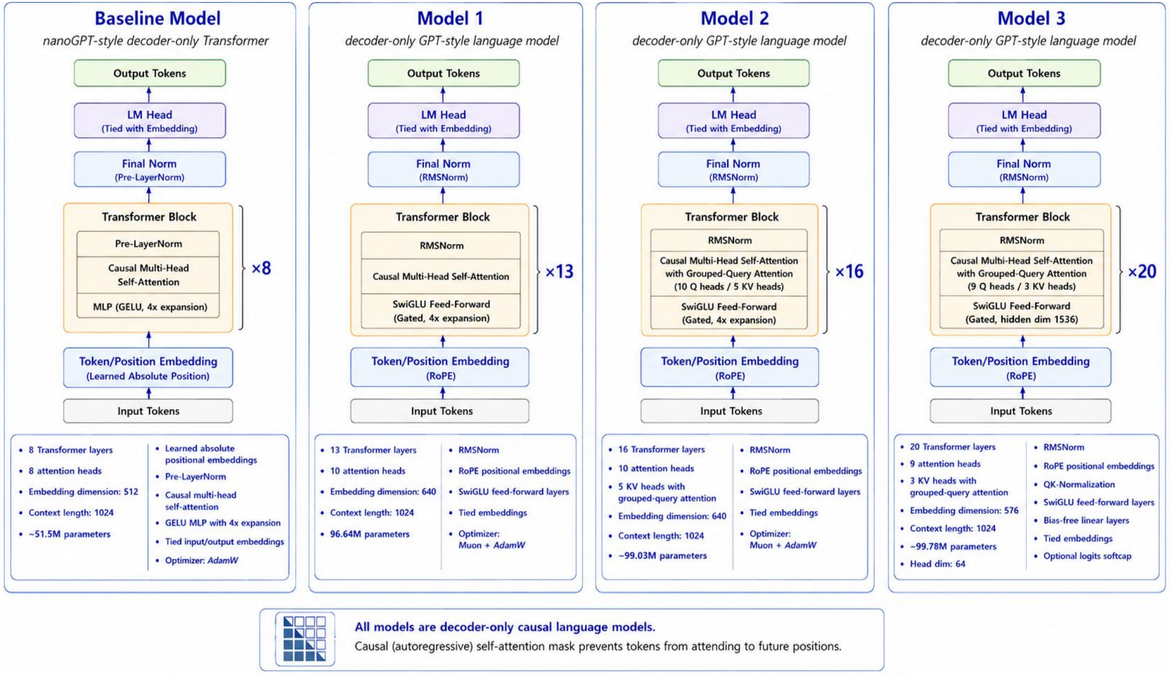
Optimal Mixture:

- FineWeb-heavy (50%)
- Wikipedia (20%)
- Science (15%)
- Books (15%)

Key Finding: Data composition is a primary driver of model perplexity.

Visualization Note: This diagram illustrates model specifications as verified by the authors. LLM assistance was used for aesthetic layout and clarity.

Architecture & Model Spec



Experiments Determining Our Final Model

Ablation studies across data source, architecture, optimizer, and mixture ratios.

1. Data Source Matters

Single-source training validation PPL at step 5k for baseline model:

- FW only: **39.67**
- OWT only: **65.21**
- FW/Wiki/Sci/Book: **38.8281**

Mixed-domain data was necessary.

2. GQA Depth Trade-off [1]

PPL at 38k for same data mix ratio (50/20/15/15):

- Model 1 (13L, no GQA): **21.3251**
- Model 2(16L, 1:2 GQA): **21.1630**
- Model 3(20L, 1:3 GQA): **20.9177**

The deeper the better for limited size

3. Optimizer Choice

PPL at 2k step comparison for Model 2:

- AdamW: **39.9728**
- Muon + AdamW: **36.5399**

Split optimizer improved efficiency.

4. FineWeb-Ratio Mix Matters For Model 1 [2]

Mixture: FineWeb / Wikipedia / Science / Books

50 / 20 / 15 / 15 (Strongest) **PPL: 21.3251**

53 / 19 / 14 / 14 (Weaker) **PPL: 21.4597**

56 / 18 / 13 / 13 (Weakest) **PPL: 21.6089**

Takeaway: Data Mix ratio will have different effects for model performance.

Conclusion: Data mixture + Model Choice + source weighting + optimizer split

[1]: Zechun Liu, Changsheng Zhao, Forrest Iandola, Chen Lai, Yuandong Tian, Igor Fedorov, Yun-yang Xiong, Ernie Chang, Yangyang Shi, Raghuraman Krishnamoorthi, Liangzhen Lai, andVikas Chandra. Mobilellm: Optimizing sub-billion parameter language models for on-device use cases. arXiv preprint arXiv:2402.14905, 2024.
[2]: Yajiao Liu, Congliang Chen, Junchi Yang, and Ruoyu Sun. Rethinking data mixture for large language models: A comprehensive survey and new perspectives. arXiv preprint arXiv:2505.21598, 2025.

5. Low-LR Fine-tuning Wins

Progressive low-LR continuation consistently improves perplexity (PPL) over longer training.

Experiment 1: Model 1 with 56 / 18 / 13 / 13 Mix

- 38k FT: 21.61
- 42k FT: 20.23
- 44k FT: 19.95
- 45k FT: 19.70
- 45.75k FT: 19.41
- 46.75k FT: 19.34
- 47.75k FT: 19.28
- 49.5k FT: 19.13
- **50.75k Best: PPL 19.09**

Experiment 2: Model 2 with 50 / 20 / 15 / 15 Mix

- 38k FT: 21.1630
- 42k FT: 19.6659
- 44k FT: 19.5598
- 50k FT: 19.4883
- 62k FT: 19.2104
- 63.6k FT: 19.0960
- 66k FT: 19.0120
- 69.2k FT: 18.9168
- **69.7k Best:PPL: 18.7681**

Takeaway: Evaluate intermediate checkpoints; the final state is not always the best performance.

Project Retrospective & Roadmap

What Have We Learned?

- Data quality and mixture mattered more than training time.
- FineWeb-heavy mix worked best; too much weakened performance.
- Low-LR fine-tuning gave consistent gains.
- Deeper over Wider for limited Parameters

Strengths & Limitations

Strengths:

- Modern sub-100M GPT-style architecture.
- Effective mixed-domain recipe (50/20/15/15).
- Strong small LR fine tuning for more steps

Limitations:

- Compute budget limited full-scale ablations.
- Manual tracking and evaluation processes.
- Broader architecture variants was not explored

Future Work Roadmap

- Automate logging, validation, and checkpoint selection.
- Compare data mixtures under controlled budgets and run pilot sweeps.
- Explore distillation and broader sub-100M architecture variants.

Strategic takeaway: Iterate on data mixtures and automate tracking to optimize sub-100M performance.

Q&A